Iris Workflow Comparison

Lucas Summers

Comparison of three classification workflows on the Iris dataset.

MODELS

MODEL 1 [BASELINE]

LogisticRegression

Description	Logistic Regression evaluated on a standard train/test split.
Type	Classification
Data	Train/Test Split
Params	22
Tuned	None

MODEL 2

RandomForestClassifier

Description	Random Forest evaluated with 5-fold stratified cross-validation.
Type	Classification
Data	5-fold CV
Params	19
Tuned	None

MODEL 3

KNeighborsClassifier

Description	KNN tuned with GridSearchCV and evaluated with 5-fold CV.
Type	Classification
Data	5-fold CV
Params	16
Tuned	2 params

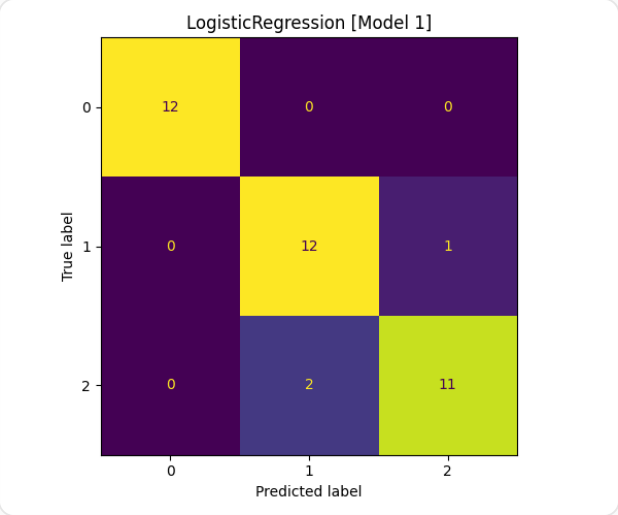
METRICS

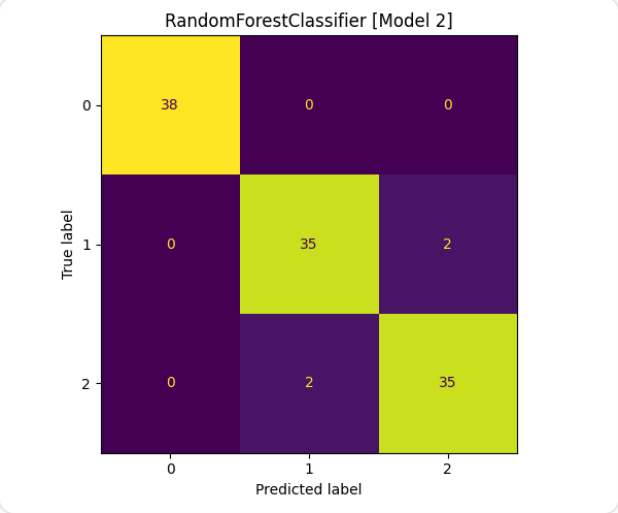
Metric	Model 1	Model 2	Model 3	Best
Accuracy	0.9211	0.9644 (+0.0434)	0.9731 (+0.0521)	Model 3
F1 Score (Macro)	0.9230	0.9629 (+0.0399)	0.9730 (+0.0500)	Model 3
F1 Score (Weighted)	0.9209	0.9638 (+0.0429)	0.9730 (+0.0520)	Model 3
Precision (Macro)	0.9246	0.9709 (+0.0463)	0.9769 (+0.0522)	Model 3
Precision (Weighted)	0.9226	0.9701 (+0.0474)	0.9762 (+0.0536)	Model 3
Recall (Macro)	0.9231	0.9619 (+0.0388)	0.9726 (+0.0495)	Model 3
Recall (Weighted)	0.9211	0.9644 (+0.0434)	0.9731 (+0.0521)	Model 3

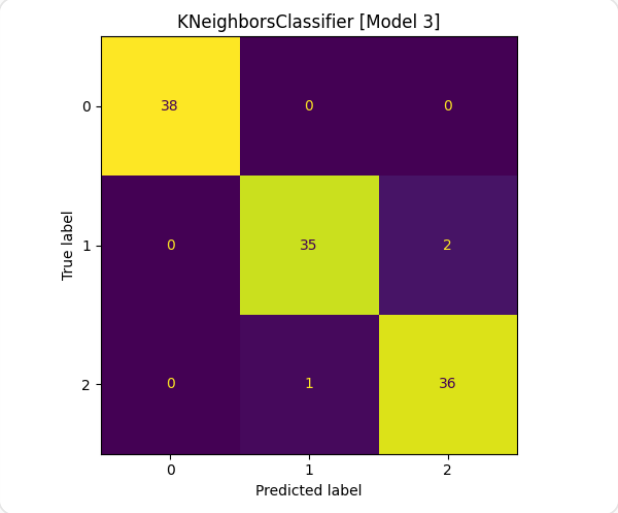
Metrics may be drawn from different evaluation splits across models.

VISUALIZATIONS

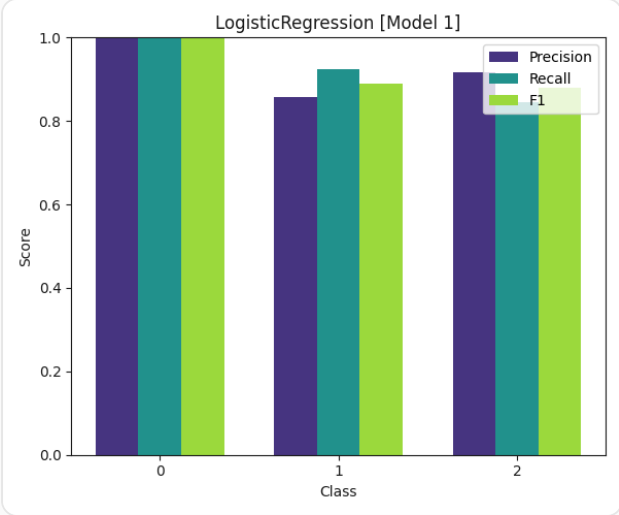
Confusion Matrix

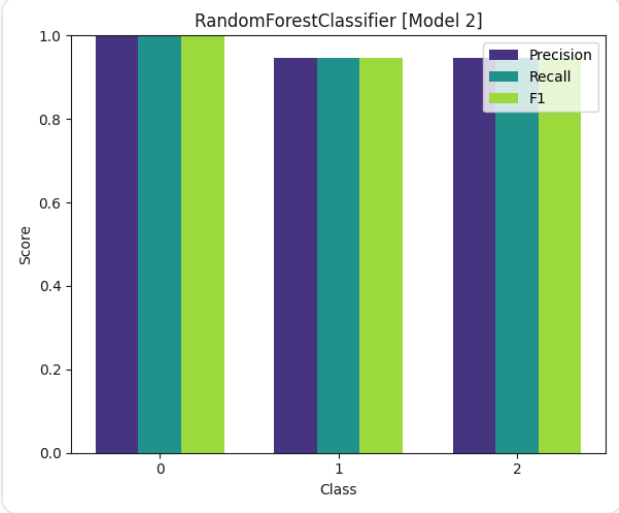


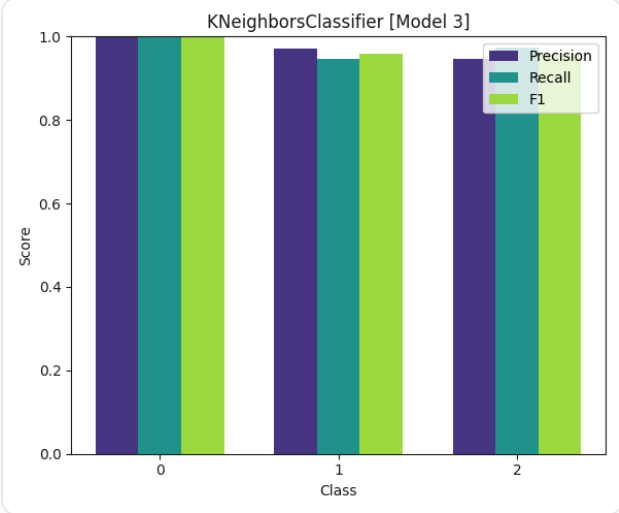




Per-Class Metrics







[Model 1] Iris Classification Report - Train/Test Split

Lucas Summers

Logistic Regression evaluated on a standard train/test split.

MODEL OVERVIEW

Name	LogisticRegression
Type	Classification
Sklearn	True (v1.8.0)
Parameter Count	22

DATASET OVERVIEW

Features	4
Total Observations	150
CV Folds	None
Train	112 (74.7%)
Test	38 (25.3%)

Class Distribution

Class	Train	Test	Overall %
0	38	12	33.3%
1	37	13	33.3%
2	37	13	33.3%

HYPERPARAMETERS

memory	None	transform_input	None
verbose	False	scaler	StandardScaler
model	LogisticRegression	scaler__copy	True
scaler__with_mean	True	scaler__with_std	True
model__C	1.0	model__class_weight	None
model__dual	False	model__fit_intercept	True
model__intercept_scaling	1	model__l1_ratio	0.0
model__max_iter	1000	model__n_jobs	None
model__penalty	deprecated	model__random_state	None
model__solver	lbfgs	model__tol	0.0001
model__verbose	0	model__warm_start	False

METRICS (TRAIN/TEST SPLIT)

Metric	Train	Test	Gap
Accuracy	0.9643	0.9211	+0.0432
F1 Score (Macro)	0.9640	0.9230	+0.0410
F1 Score (Weighted)	0.9643	0.9209	+0.0434
Precision (Macro)	0.9640	0.9246	+0.0394
Precision (Weighted)	0.9643	0.9226	+0.0417
Recall (Macro)	0.9640	0.9231	+0.0409
Recall (Weighted)	0.9643	0.9211	+0.0432

*Gap = Train - Test. Large positive values may indicate overfitting.

METRICS (PER-CLASS)

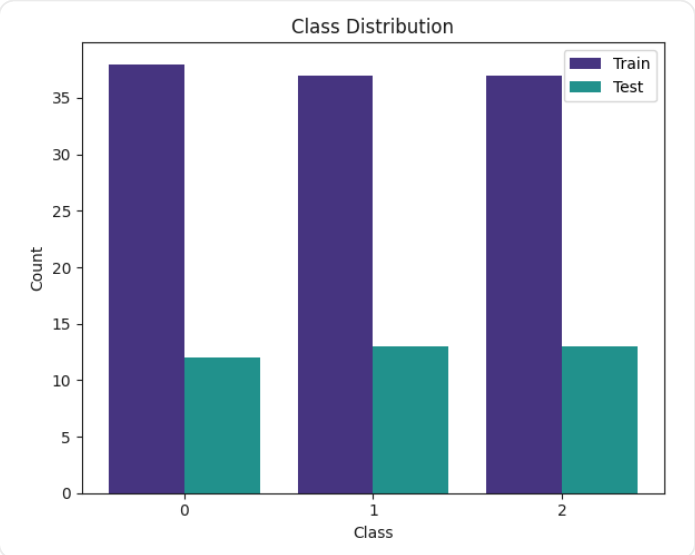
Train

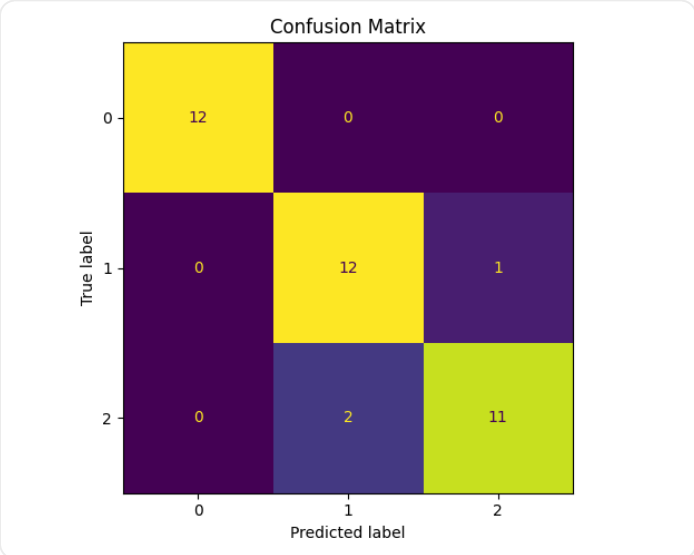
Class	Precision	Recall	F1
0	1.0000	1.0000	1.0000
1	0.9459	0.9459	0.9459
2	0.9459	0.9459	0.9459

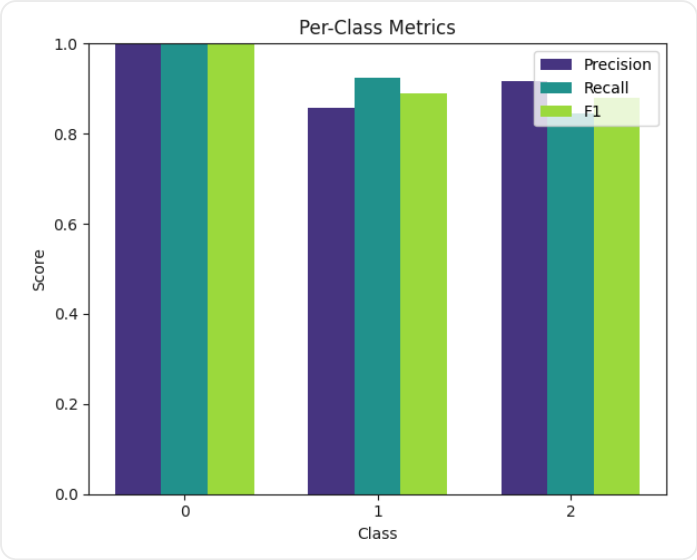
Test

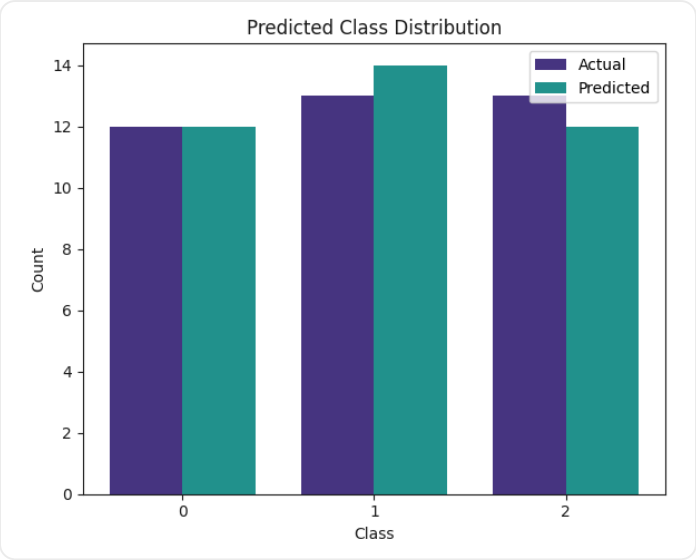
Class	Precision	Recall	F1
0	1.0000	1.0000	1.0000
1	0.8571	0.9231	0.8889
2	0.9167	0.8462	0.8800

VISUALIZATIONS






Per-Class Metrics
Precision
Recall
F1
Score
Class
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2


Predicted Class Distribution
Actual
Predicted
Count
Class
14
12
10
8
6
4
2
0
0
1
2

[Model 2] Iris Classification Report - Cross-Validation

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Random Forest evaluated with 5-fold stratified cross-validation.

MODEL OVERVIEW

Name	RandomForestClassifier
Type	Classification
Sklearn	True (v1.8.0)
Parameter Count	19

DATASET OVERVIEW

Features	4
Total Observations	112
CV Folds	5

Class Distribution

Class	CV	Overall %
0	38	33.9%
1	37	33.0%
2	37	33.0%

HYPERPARAMETERS

bootstrap	True	ccp_alpha	0.0

class_weight	None	criterion	gini
max_depth	4	max_features	sqrt
max_leaf_nodes	None	max_samples	None
min_impurity_decrease	0.0	min_samples_leaf	1
min_samples_split	2	min_weight_fraction_leaf	0.0
monotonic_cst	None	n_estimators	200
n_jobs	None	oob_score	False
random_state	42	verbose	0
warm_start	False		

METRICS (CROSS-VALIDATION)

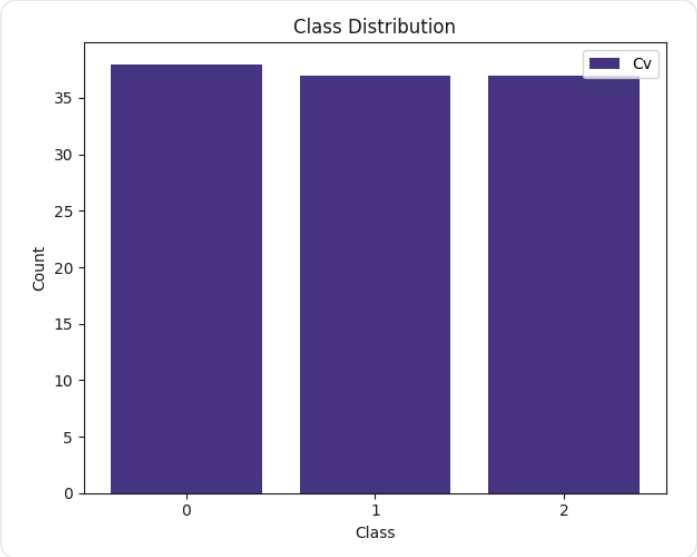
Metric	Mean	Std Dev	Min	Max
Accuracy	0.9644	0.0328	0.9130	1.0000
F1 Score (Macro)	0.9629	0.0347	0.9074	1.0000
F1 Score (Weighted)	0.9638	0.0336	0.9106	1.0000
Precision (Macro)	0.9709	0.0258	0.9333	1.0000
Precision (Weighted)	0.9701	0.0267	0.9304	1.0000
Recall (Macro)	0.9619	0.0356	0.9048	1.0000
Recall (Weighted)	0.9644	0.0328	0.9130	1.0000

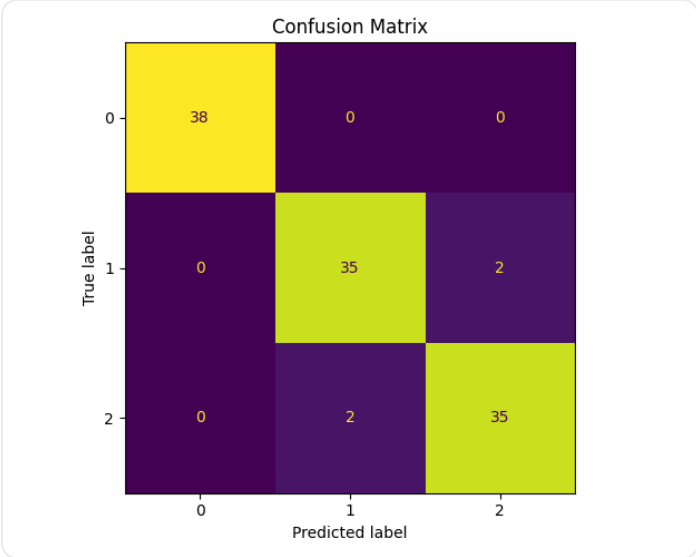
METRICS (PER-CLASS)

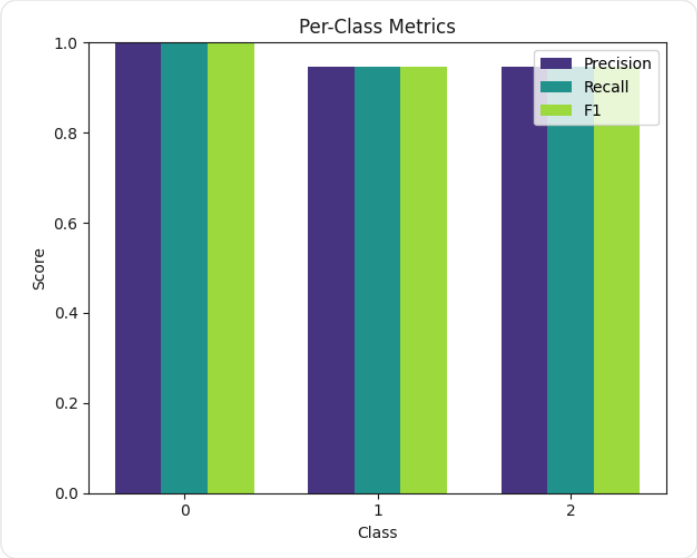
Class	Precision	Recall	F1

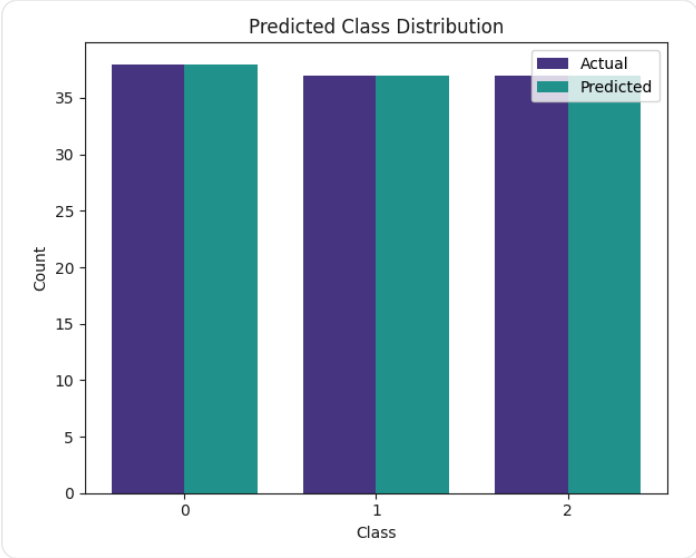
0	1.0000	1.0000	1.0000
1	0.9459	0.9459	0.9459
2	0.9459	0.9459	0.9459

VISUALIZATIONS









[Model 3] Iris Classification Report - Tuning + Cross-Validation

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KNN tuned with GridSearchCV and evaluated with 5-fold CV.

MODEL OVERVIEW

Name	KNeighborsClassifier
Type	Classification
Sklearn	True (v1.8.0)
Parameter Count	16

DATASET OVERVIEW

Features	4
Total Observations	112
CV Folds	5

Class Distribution

Class	CV	Overall %
0	38	33.9%
1	37	33.0%
2	37	33.0%

HYPERPARAMETERS

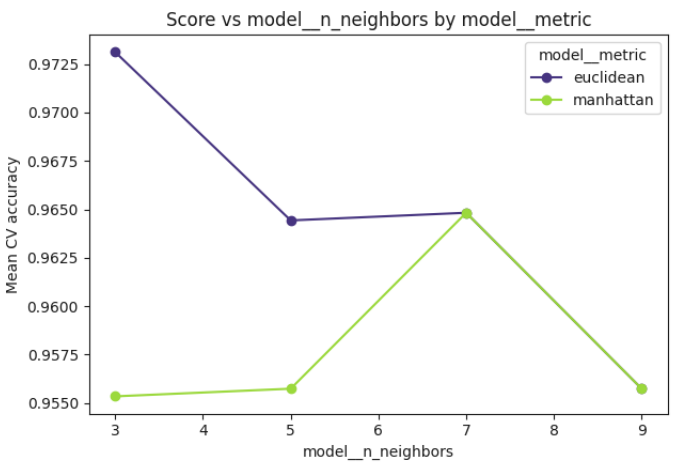
memory	None	transform_input	None

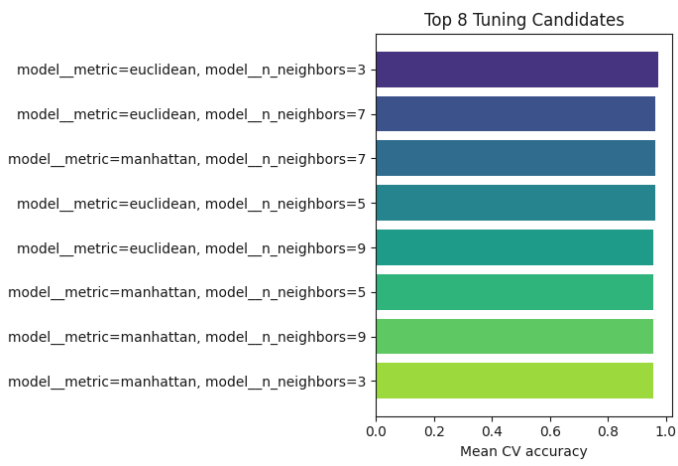
verbose	False	scaler	StandardScaler
model	KNeighborsClassifier	scaler__copy	True
scaler__with_mean	True	scaler__with_std	True
model__algorithm	auto	model__leaf_size	30
model__metric	euclidean	model__metric_params	None
model__n_jobs	None	**model__n_neighbors**	3
model__p	2	model__weights	uniform

Tuning Summary

Method	GridSearchCV
Metric	accuracy
CV Folds	5
Candidates	8
Best Score	0.9731

Tuning Plots

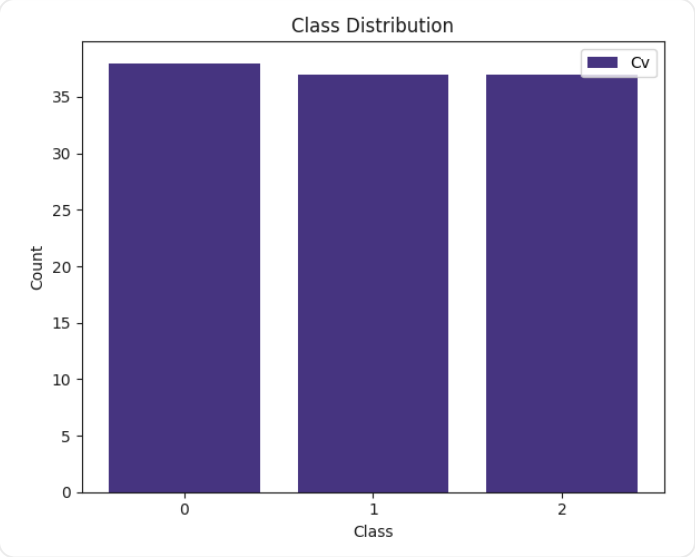
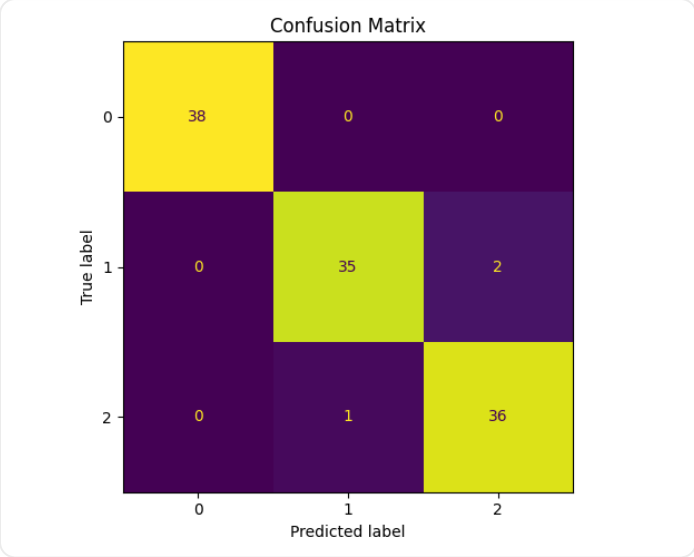
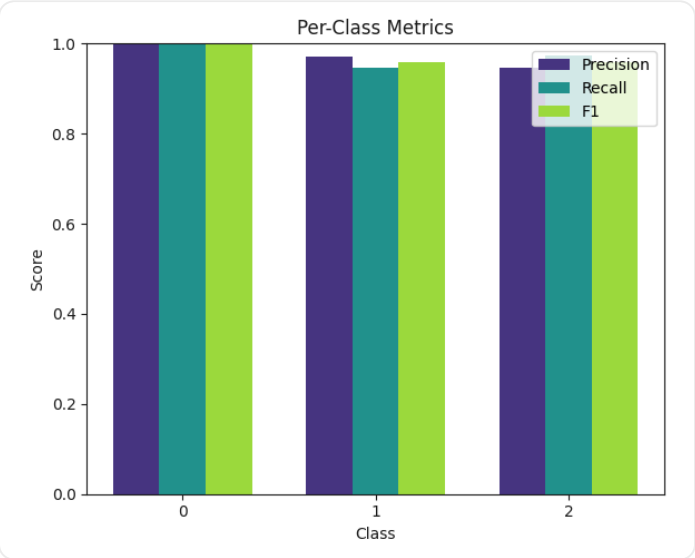
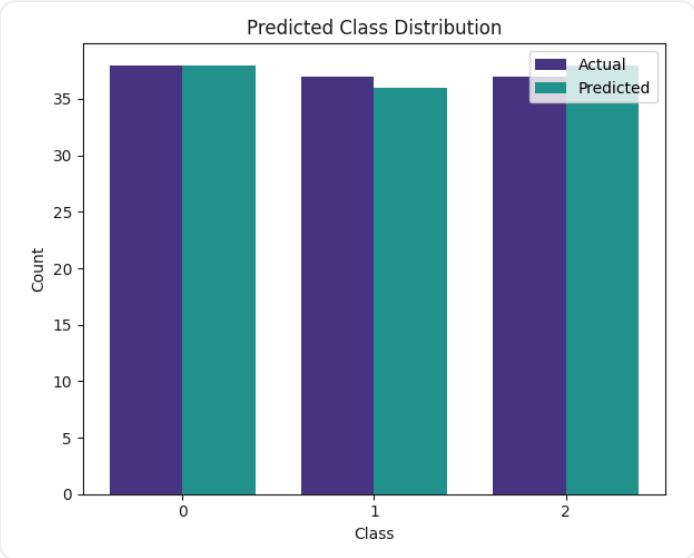




METRICS (CROSS-VALIDATION)

Metric	Mean	Std Dev	Min	Max
Accuracy	0.9731	0.0220	0.9545	1.0000
F1 Score (Macro)	0.9730	0.0220	0.9548	1.0000
F1 Score (Weighted)	0.9730	0.0221	0.9541	1.0000
Precision (Macro)	0.9769	0.0190	0.9583	1.0000
Precision (Weighted)	0.9762	0.0194	0.9596	1.0000
Recall (Macro)	0.9726	0.0225	0.9524	1.0000
Recall (Weighted)	0.9731	0.0220	0.9545	1.0000

METRICS (PER-CLASS)

Class	Precision	Recall	F1
0	1.0000	1.0000	1.0000
1	0.9722	0.9459	0.9589
2	0.9474	0.9730	0.9600

VISUALIZATIONS